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The Atomization Fallacy: Why Flat Graphs Fail Composite Systems and Neuro-Symbolic AI

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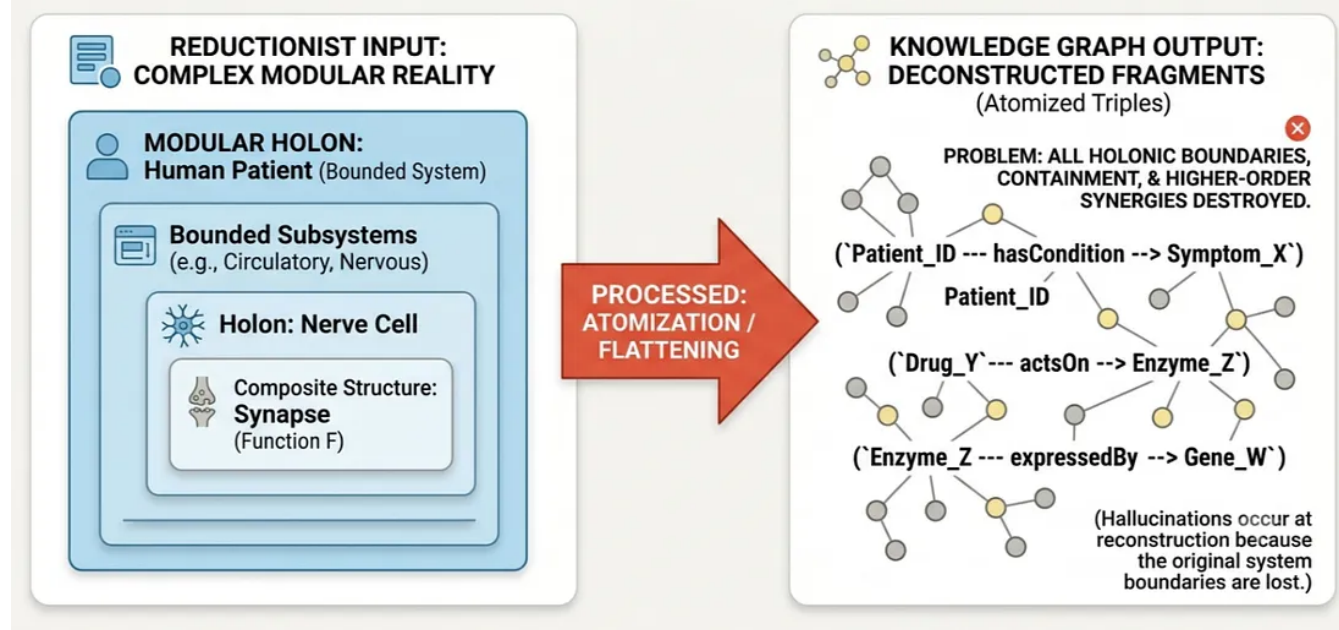


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UNVEILING THE ATOMIZATION FALLACY: WHEN FLAT GRAPHS FAIL COMPLEX SYSTEMS



Abstract

The current artificial intelligence landscape presents Knowledge Graphs (KGs) and Graph Retrieval-Augmented Generation (GraphRAG) as deterministic “sources of truth” designed to eliminate Large Language Model (LLM) hallucinations and bridge

neuro-symbolic reasoning. However, this paradigm conceals a profound structural fallacy: reducing complex, composite systems into flat dyadic graphs or atomic triples forces severe semantic loss, destroys boundaries, and atomizes context.

By examining graph expressiveness ceilings, formal mereotopology, the failure of syntactic “workarounds” (RDF*, hypergraphs, SysML v2), the holonic paradox, and the systems engineering contributions of Dr. Nicolas Figay, this paper demonstrates why flat graphs fail composite systems — and outlines what a true neuro-symbolic substrate must look like.

1. The Reductionist Trap: Atomization in Modern Knowledge Representation

To make unstructured human knowledge machine-readable, mainstream computer science routinely applies a reductionist transformation: decomposing complex, multi-scale realities into binary relationships:

(Subject) —[Predicate]—> (Object)

While this model excels at indexing simple binary relationships (e.g., Alice —knows—> Bob), it breaks down when applied to composite, modular systems — such as aircraft architectures, biological pathways, legal frameworks, or manufacturing lifecycles.

When a multi-entity interaction or nested subsystem is forced into a flat graph, the network loses its containment boundaries, higher-order synergy, and context-dependent state. The resulting graph is not a faithful digital twin of reality, but an atomized mesh of disconnected fragments.

2. Mathematical and Topological Limits of Dyadic Graphing

A. Loss of Non-Additive n-Body Synergy

In standard graph theory, an edge is defined strictly as a dyadic set $E \subseteq V \times V$. When a composite system exhibits a 3-body or n-body interaction (e.g., three co-prescribed drugs producing a unique, non-additive side effect), projecting this system into a graph requires creating a clique of pairwise edges:

$$C_n = \{ (v_i, v_j) \mid 1 \leq i < j \leq n \}$$

This projection causes structural ambiguity: a pairwise graph cannot distinguish between a single, irreducible n -body composite event and $n(n - 1) / 2$ independent pairwise interactions.

B. The 1-WL Expressiveness Bound in Graph Neural Networks (GNNs)

Message Passing Graph Neural Networks (MPNNs) aggregate neighbor features via pairwise edges:

$$h_v^{(k+1)} = \text{UPDATE}(h_v^{(k)}, \text{AGGREGATE}(\{ h_u^{(k)} \mid u \in N(v) \}))$$

This mechanism is mathematically bounded by the 1st-order Weisfeiler–Leman (1-WL) graph isomorphism test. MPNNs are blind to higher-order substructures (e.g., distinguishing between independent triangles versus cliques) and struggle to learn composite compositionality without explicit topological extensions (such as simplicial or cell complexes).

3. Deconstructing Modern Syntactic Workarounds vs. Formal Mereotopology

Database engineers and standards bodies have introduced several mechanisms to bypass flat graph limits. However, a rigorous analysis demonstrates that these solutions are often syntactic workarounds that leave the underlying atomization intact.

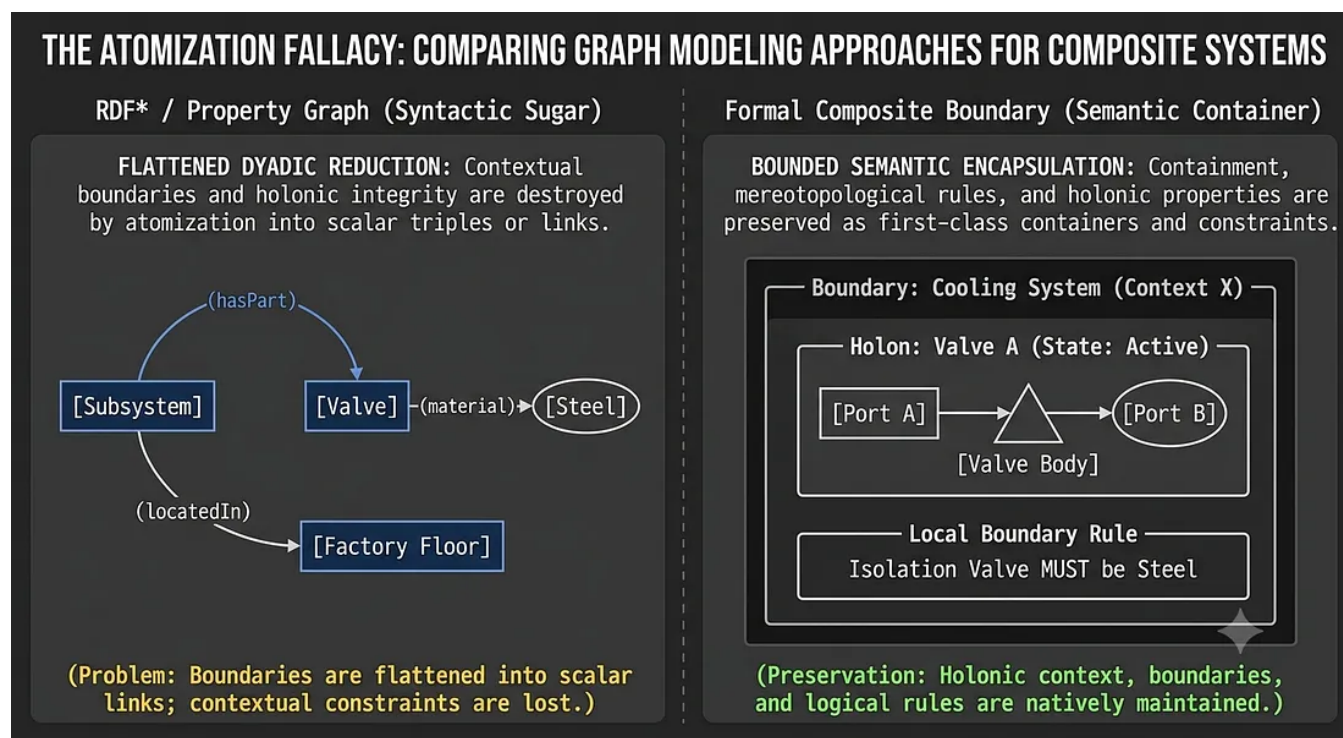
A. Syntactic Workarounds

- **RDF* / RDF 1.2:** Allows triples to be nested inside triples $\langle\langle s \ p \ o \rangle\rangle \ p_2 \ o_2$. While convenient for attaching edge metadata (provenance, confidence), it remains bound to first-order dyadic logic. It attaches annotations to edges but fails to create first-class, spatial or semantic containment boundaries.
- **TypeDB & Hypergraphs:** Allows n -ary hyperedges connecting arbitrary node sets. However, hypergraph engines group data *syntactically* (like XML) without enforcing *semantic grouping* (like XML Schema). Without domain-level

mereotopology, a hyperedge cannot distinguish an ad-hoc cluster of attributes from an encapsulated, composite object.

- UML / SysML v1 & v2: Uses composition relationships, internal block diagrams, and KerML namespaces. However, SysML treats parts as *properties or usages* of a parent classifier rather than autonomous entities, stripping parts of their independent identity when evaluated outside the parent context.

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B. Formal Mereotopology: Giancarlo Guizzardi's Unified Foundational Ontology (UFO)

Formal mereotopology combines part-whole relations (mereology) with boundary/topological relations (topology). Guizzardi's UFO demonstrates why simple graph properties fail:

1. Non-Transitivity of Functional Parts: Simple graph edges assume transitivity (A partOf B and B partOf C $\Rightarrow$ A partOf C). Mereotopology proves that *functional composition* is non-transitive: a handle is part of a door, and a door is part of a house, but a handle is not a functional part of a house. Flat graphs flatten these distinct relation types into generic partOf edges.

- Sortal Integrity and Anti-Rigidity: A composite object requires a *sortal classifier* that supplies a principle of identity. Flat graph nodes lack sortal boundaries, leading to semantic collapse when a node changes state across lifecycles.

4. The Holon Paradox & Boundary Erosion Under Virtualization

Arthur Koestler introduced the concept of a holon — an entity that is simultaneously an autonomous individual *whole* and a constituent *part* of a larger system.

The Logical Paradox in Standard Formalisms

Standard Knowledge Representation languages enforce a strict dichotomy between:

- Individuals (A-Box): Concrete entity instances.
- Classes/Sets (T-Box): Abstract category definitions.

A holon breaks this separation:

- At Level N (macro context), the holon must act as an atomic individual node with single-point properties.
- At Level N — 1 (micro context), the same holon must act as a class/container enclosing a complex internal graph.

First-Order Logic (FOL) and Description Logic (OWL) engines cannot model an entity that is simultaneously an atomic node and a container set without incurring decidability failures or type-collapse errors.

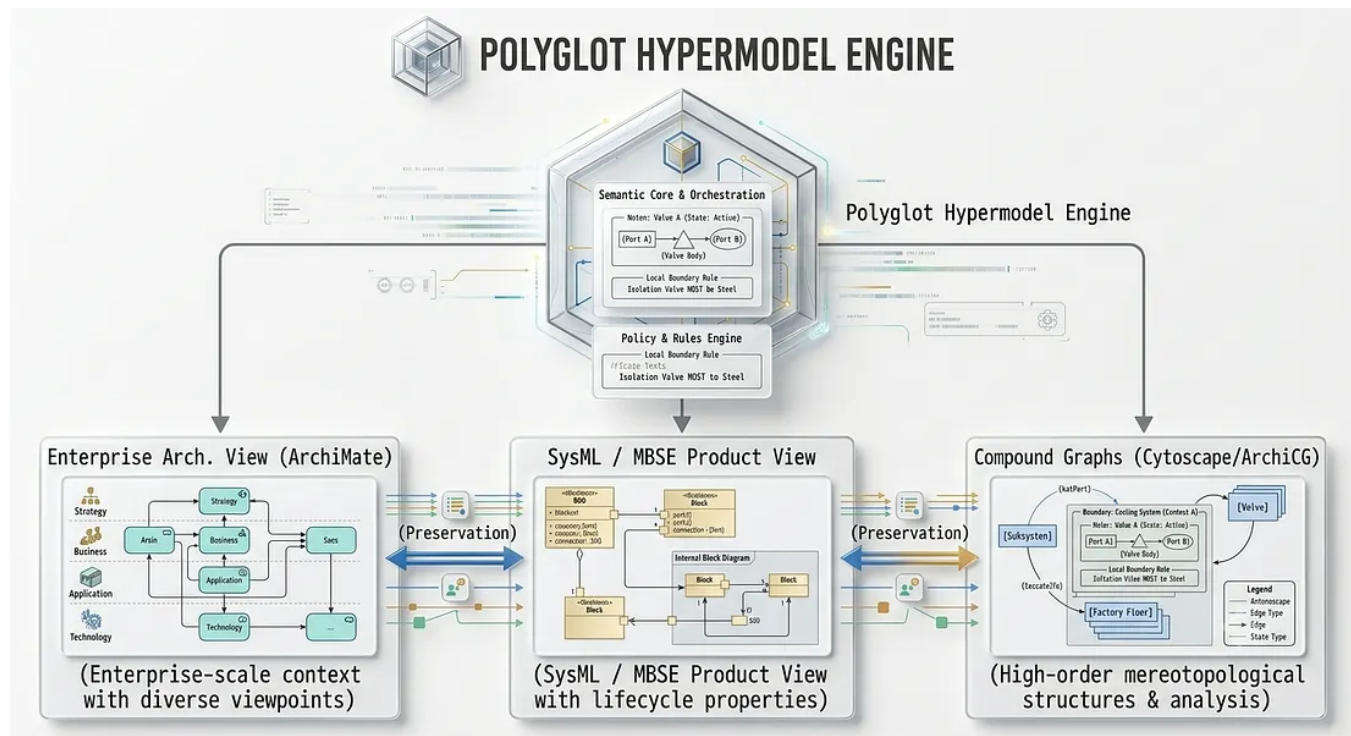
Boundary Erosion Under Virtualization

When physical or software systems are virtualized across distributed platforms (e.g., IoT digital twins, cloud PLM), single-formalism graph mappings dissolve encapsulation boundaries. What was an isolated, governed subsystem collapses into a flat mesh of cross-layer relationships, exposing internal variables to unauthorized contexts and destroying configuration integrity.

5. Applied Architecture Solution: Polyglot Hypermodels?

While computer science seeks a single theoretical “super-formalism,” my research have been addressing the problem through multi-representation orchestration.

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This research in Product Lifecycle Management (PLM) and Model-Based Systems Engineering (MBSE) aims at proposing key innovations:

1. **Shift to Composite Models:** Rejecting flat semantic graphs in favor of composite models that explicitly preserve component containment boundaries, contextual dependencies, and hierarchical decomposition.
2. **Polyglot Hypermodels & Compound Graphs (ArchiMateCG):** Instead of compressing all domain knowledge into a single language (such as OWL or SysML), Figay developed polyglot hypermodels. These frameworks orchestrate *multiple domain-specific modeling languages simultaneously*, leveraging compound graphs (where nodes enclose subgraphs) to support interactive nested visualization and semantic alignment across viewpoints.
3. **Semantic Preservation across Virtualization:** Enforcing meta-model governance that preserves boundary semantics when product data is virtualized across extended enterprise networks.

6. Implications for Neuro-Symbolic AI and GraphRAG

The prevailing enthusiasm surrounding Knowledge Graphs and GraphRAG in Generative AI routinely overlooks this atomization problem.

When unstructured knowledge is ingested into a standard Property Graph or RDF store to ground LLMs, the retrieval pipeline suffers from Context Fragmentation:

Flat GraphRAG Ingestion (Lossy):

[Subsystem] —(hasPart)—> [Valve] —(material)—> [Steel]

(Result: Context shredded into scalar paths; containment boundaries destroyed.)

Bounded Hypermodel Ingestion (Context Preserved):

HOLON / CONTAINER: Primary Cooling Subsystem

- Lifecycle State: Virtualized / Active
- Boundary Rule: Isolation Valve MUST be Steel

Internal Topology: [Valve A] <---> [Sensor B]

Three Modes of GraphRAG Failure

1. **Shredding at Ingestion:** Rich narrative context and composite system dynamics are atomized into scalar, dyadic edges during graph construction.
2. **Blind Traversal:** When GraphRAG traverses a graph path (Component A — partOf → Subsystem B — locatedIn → Factory C), it traverses relationships without knowing which logical boundary it crossed, whether an entity is acting as a whole or a part, or what emergent whole-system constraints apply.
3. **Hallucination at Reconstruction:** Because the graph engine destroyed encapsulation boundaries during ingestion, the downstream LLM is forced to “guess” how the atomic fragments fit back together — re-introducing hallucinations at the synthesis stage.

Conclusion: Toward Bounded, Multi-Representation AI

If the AI community intends to pair LLMs with symbolic structures to achieve true neuro-symbolic reasoning, it must move past the assumption that flat graphs faithfully represent reality.

Knowledge substrates for AI must evolve from flat triple meshes toward topologically aware, polyglot hypermodel architectures. Only by natively preserving containment boundaries, holonic scale, and multi-view representation can we bridge the gap between statistical generative AI and true explanatory, deterministic systems modeling.

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Written by Dr Nicolas Figay

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